



Thesis Project:

Performance Comparison of Various Loss Functions in Benchmark Datasets

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Introduction

Loss functions are the core components of machine learning that guide the optimization process. Loss functions measure model performance by calculating the deviation of a model's predictions from the "actual" predictions.

Motivation

- Optimization Impact:** The choice of loss function directly dictates convergence speed, stability, and generalization.
- Domain Sensitivity:** Performance varies significantly across tabular, image, text, time series, and audio domains.
- Robustness:** Identifying specific loss functions that excel under class imbalance or label noise is critical for real-world applications.

Objectives

- Systematic Evaluation:** Conduct a unified comparison of common loss functions across multiple benchmark datasets.
- Performance Metrics:** Analyze influence on model accuracy, convergence behavior, and robustness.
- Optimization Guidelines:** Identify the most suitable loss functions for specific data characteristics (e.g., noisy or imbalanced data).

Methodology

The research follows a multimodal experimental process to evaluate the performance of loss functions in various data domains:

- Multimodal Data Integration:** The study includes benchmark datasets from five different domains (Table, Image, Text, Time Series, and Audio) and provides a comprehensive evaluation of loss functions under varying feature complexities.
- Controlled Preprocessing:** Each dataset has been subjected to domain-specific optimization, including mean/mode imputation, one-hot encoding, and standard scaling. Stratified splitting has been maintained to preserve class distributions.
- Architectural Diversity:** Experiments have been conducted on a wide range of models, from low-capacity linear models (Logistic Regression) to high-capacity nonlinear deep neural networks (MLP, CNN, 1D-CNN, 2D-CNN).
- Experimental Rigor:** To ensure methodological validity, tree-based models were excluded as they do not have differentiable optimization goals. All gradient-based models used fixed hyperparameters (Adam optimization algorithm, fixed learning rate) to isolate the effect of the loss function.
- Comparative Framework:** Performance was compared using robustness and convergence metrics, including Accuracy, Precision, Recall, MAPE, RMSE, F1-Score, ROC-AUC, and Loss Curves, with a particular focus on addressing class imbalance and label noise.

Key Observations on Loss Function Performance

1. Dynamics in Classification Tasks

- Tabular Data & Class Imbalance:** In structured datasets like Titanic and Breast Cancer, while BCE and Hinge Loss yielded comparable results on balanced subsets, Weighted BCE (WBCE) and Focal Loss (FL) significantly enhanced the model's discriminative power in imbalanced scenarios by prioritizing minority class representation.
- Computer Vision & Audio (Unstructured Data):** Standard Cross-Entropy (CE) demonstrated robust baseline performance across Fashion-MNIST, CIFAR-10, and UrbanSound8K. However, the integration of Label Smoothing (LS) mitigated model overconfidence, thereby improving generalization. Furthermore, Focal Loss proved superior in handling noisy or hard-to-classify samples, particularly within the audio domain.

2. Regression and Continuous Data Analysis

- Sensitivity to Outliers:** In regression tasks such as California Housing and Wine Quality, MSE exhibited high sensitivity to outliers due to its quadratic penalty. Conversely, MAE demonstrated greater robustness, albeit with slower convergence.
- Hybrid Optimization:** Huber Loss and Log-Cosh (LC) effectively bridged the gap between MSE's differentiability and MAE's robustness. In time-series analysis (FordA), Log-Cosh ensured a more stable training trajectory by preventing gradient explosions during optimization.

3. NLP and Margin-Based Approaches

- Text Classification:** For NLP tasks like IMDB and SMS Spam, margin-based approaches (Margin Loss, Hinge Loss) achieved high accuracy by maximizing class separability. While regression losses (MSE/MAE) were evaluated, probabilistic (BCE) and margin-based functions remained more compatible with the high-dimensional sparsity of text data.

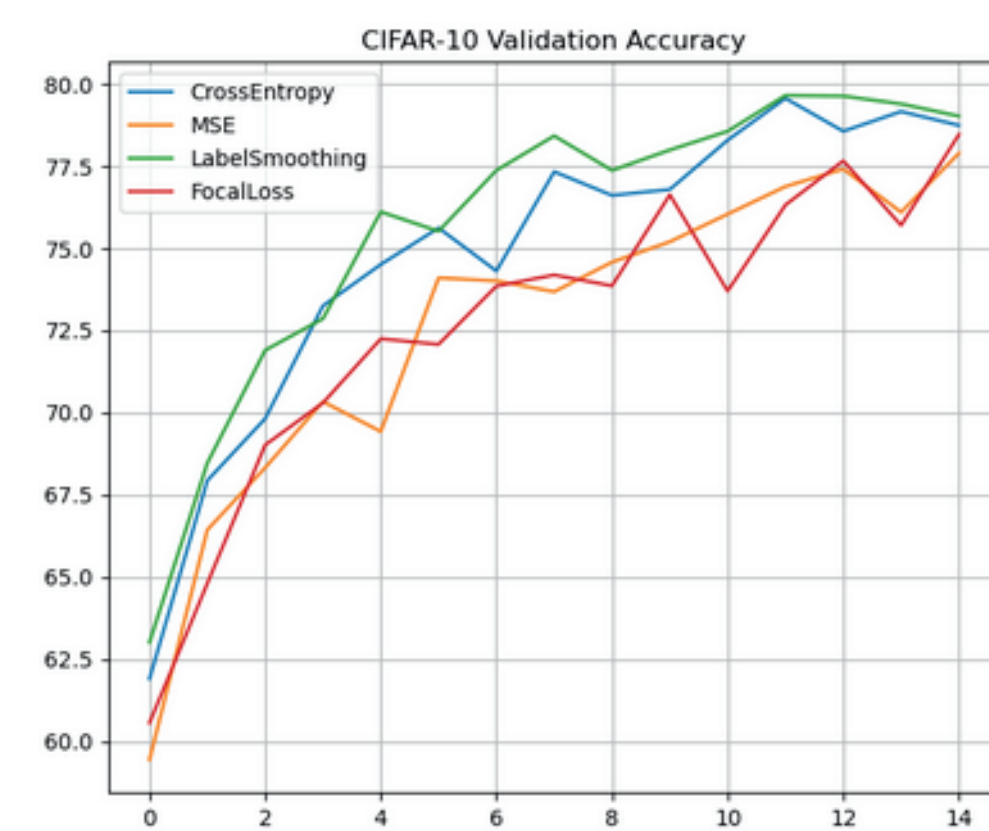
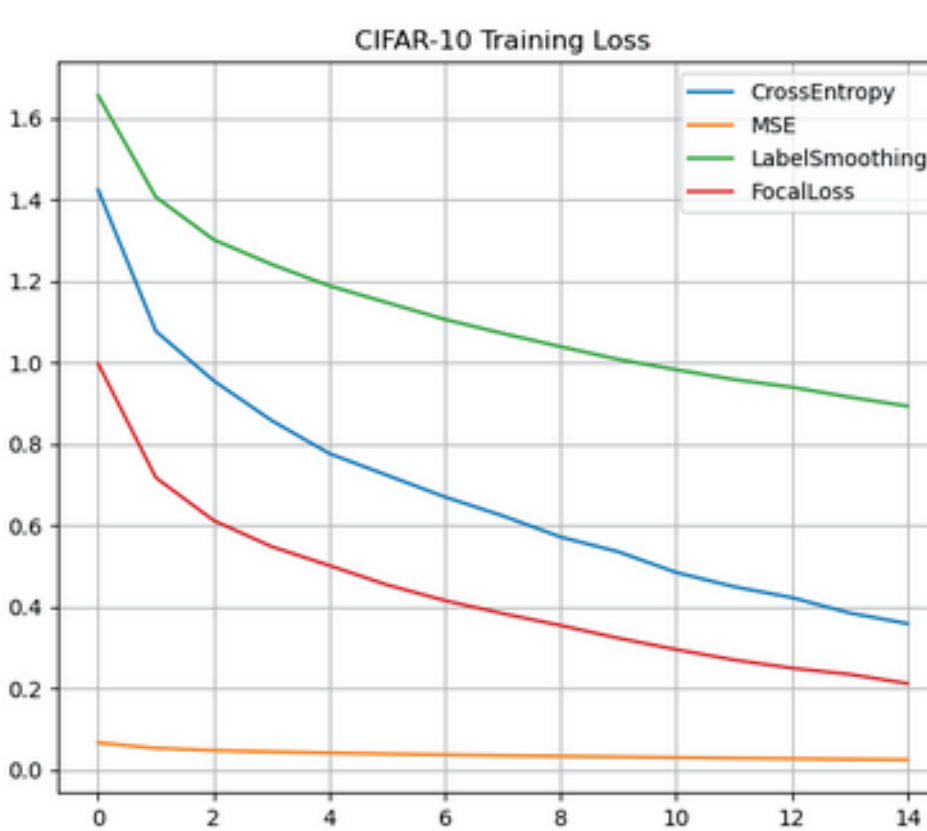
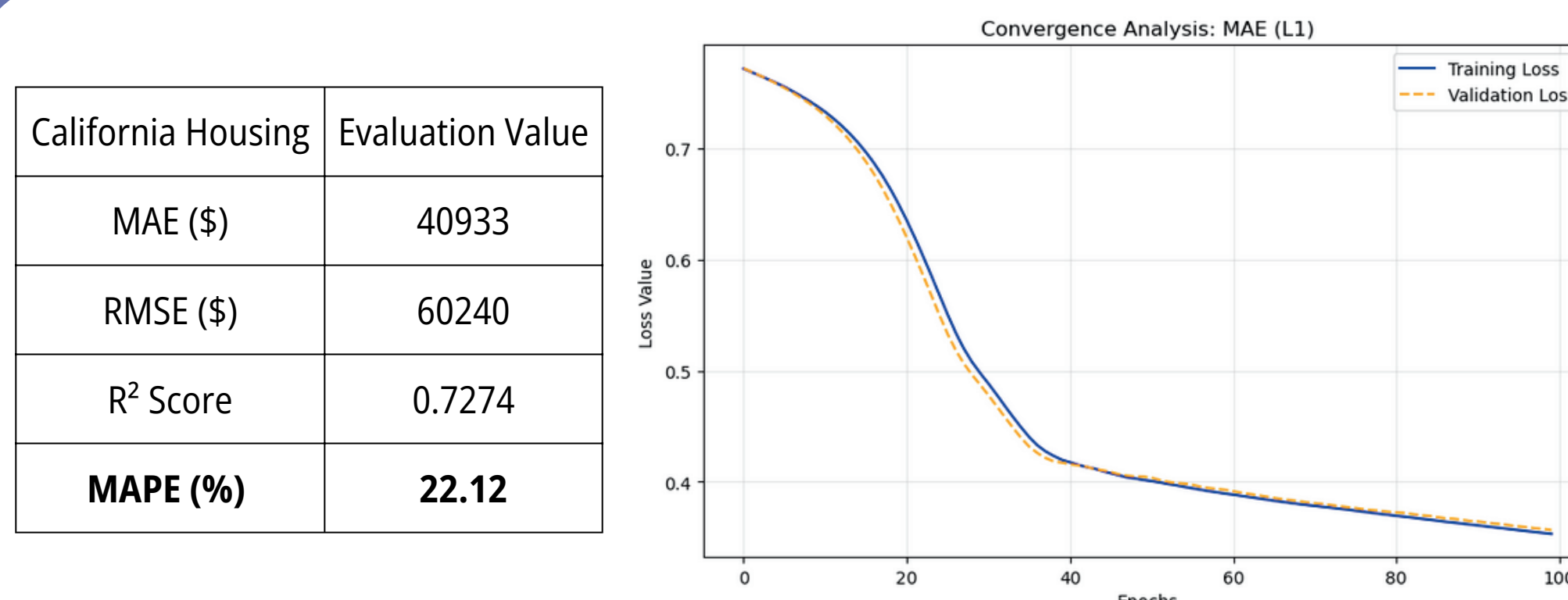
Dataset	Type	Used Models	Loss Functions
Titanic	Tabular	LR ,MLP	BCE, WBCE, Focal , Hinge
Breast Cancer	Tabular	LR ,MLP	BCE, WBCE , Focal, Hinge
Fashion-MNIST	Image	CNN	CE, MSE, LS , FL
CIFAR-10	Image	CNN	CE, MSE, LS , FL
IMDB	Text	LR ,MLP	BCE, MSE, MAE, Huber , Margin
SMS Spam	Text	LR ,MLP	BCE , MSE , MAE, Huber, Margin
California Housing	Regression	MLP	MSE, MAE , Huber, LC
Wine Quality	Regression	MLP	MSE , MAE, Huber, LC
FordA	Time Series	1D-CNN	LC , MSE, FL, CE, LS
UrbanSound8K	Audio	2D-CNN	CE , FL, LS



*BCE: Binary Cross-Entropy (Log Loss), WBCE: Weighted Binary Cross-Entropy, MSE: Mean Squared Error, CE: Cross-Entropy, LS: Label Smoothing, FL: Focal Loss, MAE: Mean Absolute Error, LC: Log-Cosh Loss



*The loss functions and models that yielded the best results in the experiments are listed in the table in bold.



Loss Function CIFAR-10	Accuracy	Weighted Avg F1
Cross-Entropy	0.929	0.928
MSE	0.927	0.928
Label Smoothing	0.932	0.932
Focal Loss	0.926	0.926

Label Smoothing CIFAR-10

Epoch [1/15] Loss: 0.9222 Acc: 88.53%
Epoch [2/15] Loss: 0.8063 Acc: 90.51%
Epoch [3/15] Loss: 0.7646 Acc: 91.13%
Epoch [4/15] Loss: 0.7389 Acc: 91.49%
Epoch [5/15] Loss: 0.7226 Acc: 92.03%
Epoch [6/15] Loss: 0.7087 Acc: 92.13%
Epoch [7/15] Loss: 0.6921 Acc: 92.58%
Epoch [8/15] Loss: 0.6775 Acc: 92.12%
Epoch [9/15] Loss: 0.6680 Acc: 92.79%
Epoch [10/15] Loss: 0.6573 Acc: 92.38%
Epoch [11/15] Loss: 0.6437 Acc: 92.67%
Epoch [12/15] Loss: 0.6350 Acc: 92.94%
Epoch [13/15] Loss: 0.6259 Acc: 92.16%
Epoch [14/15] Loss: 0.6209 Acc: 92.61%
Epoch [15/15] Loss: 0.6144 Acc: 92.79%



Conclusion

This comprehensive study demonstrates that the selection of an optimal loss function is inherently tied to data modality and class distribution. Our experimental results validate that while standard functions like Cross-Entropy and MSE provide reliable baselines, specialized losses such as Focal Loss and Log-Cosh offer superior performance in handling class imbalance and outliers, respectively. Furthermore, the integration of Label Smoothing and Hybrid Regression Losses significantly enhances model generalization across unstructured data types (Image, Audio, and Text). Ultimately, these findings underscore that a context-aware loss function strategy is critical for developing robust, high-performance deep learning models across diverse real-world benchmarks.

